

POWERFACTORY

PowerFactory 2024

Technical Reference

Current Transformer

StaCt,StaCtcore,TypCt,TypCtcore

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

This document describes the *PowerFactory* current transformers and their calculation models.

1.1 Current Transformer Type

The current transformer type (*TypCt*) facilitates the input of data for generic current transformers or from a specific data sheet.

The *Basic Data* page allows the input of one or more primary and secondary ratings for the current transformer.

The *Advanced Data* page is used to specify the accuracy class and the thermal rating. Based on the selected accuracy input option, either the rated apparent power, S_r, the rated burden, Z_r, or the voltage at accuracy limit, V_{max}, can be entered. The remaining values are calculated as follows:

Note: The list entries provided in the dialogue are only intended as a guide. Any non-zero value can be directly entered.

IEC - Apparent Power

$$Z_r = \frac{S_r}{I_{r,s}^2} \tag{1}$$

$$V_{max} = \frac{S_r \cdot ALF}{I_{r,s}} \tag{2}$$

ANSI (C) - Burden

$$S_r = Z_r \cdot I_{r,s}^2 \tag{3}$$

$$V_{max} = Z_r \cdot ALF \cdot I_{r,s} \tag{4}$$

ANSI (C) - Voltage

$$S_r = \frac{V_{max} \cdot I_{r,s}}{ALF} \tag{5}$$

$$Z_r = \frac{V_{max}}{ALF \cdot I_{r,s}} \tag{6}$$

where:

S_r is the rated apparent power in VA (*Snom*)

$I_{r,s}$ is the rated secondary current in A

Z_r is the rated burden in Ω (*Zb*)

V_{max} is the voltage at the accuracy limit in V (*Vmax*)

ALF is the accuracy limit factor (*aclimit*)

As the type itself does not specify a singular rated secondary current, the equations above will always use the first entry in the secondary tap list from the *Basic Data* page.

1.2 Current Transformer Core Type

The current transformer core type (*TypCtcore*) can be used to model a detailed current transformer core. In contrast to the *TypCt*, only one primary and secondary rating can be entered. However, it is possible to enter multiple ratios to model cores with more than one tap. The accuracy class input on the *Advanced Data* page is available per ratio and uses the equations described in Section 1.1.

1.3 Current Transformer

A current transformer (*StaCt*) can be defined without a type, with a *TypCt* or with a *TypCtcore*. Current transformers without a type are always considered as an ideal transformer with a $1:1$ ratio.

Measurement Location:
Typically, a current transformer is placed inside a cubicle (*StaCubic*) within the network model to indicate the measurement location. For advanced applications, it is also possible to specify the measurement location directly via the "Reference (*pReference*)" selection.

The following measurement locations can be selected:

- Cubicle (*StaCubic*): measure the current at the given cubicle
- Switch (*StaSwitch*): measure the current at the cubicle the given switch is stored in
- Breaker (*ElmCoup*): measure the current at *Bus i* of the given breaker
- 3-Winding Transformer (*ElmTr3*) or 4-Winding Transformer (*ElmTr4*): measure the circulating current in the delta winding; the current transformer must be stored in the cubicle associated with the delta winding
- Current Transformer (*StaCt*): measure as an auxiliary current transformer to the given current transformer

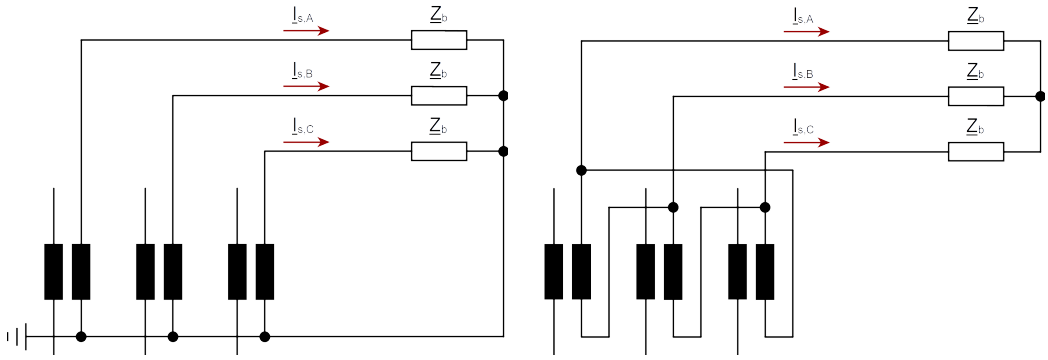
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Note: It is also possible to measure currents of DC equipment. In this case, the current transformer will always be considered as an ideal single-phase transformer.

Tap Selection:
If the selected type defines multiple primary taps, secondary taps or ratios, the selection can be made here. If multiple primary taps or ratios are available, it is possible to let the current transformer search for the primary tap or ratio which is close to, but higher than, the nominal current of the connected element by pressing the "Detect primary tap" button.

Connection:
The current transformer can be configured to measure 1-3 phases. In the case of a 3-phase current transformer, the vector group and phase rotation for the connection of the secondary side can be configured. The *wye* ("Y") and *delta* ("D") connections are shown in Figures 1.1a and 1.1b, respectively.

For 2-phase current transformers, the two measured phases (which are always *wye* connected) can be selected. For single-phase transformers, two additional phases can be selected: 'I0' and 'N'. Both will either calculate the zero sequence current from the phase currents or, if present, measure the current in the neutral wire (see Section 2.4). For certain measurement locations, such as 3W-transformer delta windings, the phase selection is fixed and cannot be changed.



(a) Wye connection (Y) (b) Delta connection (D)

Figure 1.1: 3-phase current transformer connections

The *Orientation* determines which direction of current flow is considered "positive". If the current transformer is used to measure inside a substation, the orientation is relative to the bay, regardless of the location of the transformer within the bay.

Note: This setting is inversely correlated to the grounding input commonly found in relay settings (for example). A current transformer oriented towards the connected branch ($--> Branch$) is grounded on the busbar side and vice versa.

Additional Cores:
It is possible to group several cores (see Section 1.4) under one current transformer. They can be created and accessed using the buttons "Add Core" and "Edit Cores". The contained cores are listed with their nameplate.

Additional Data:
The *Additional Data* page can be used to enable and configure saturation for the model. For detailed information, please see Section 2.3 for 2- or 3-phase current transformers and Section 2.5 for single-phase current transformers.

1.4 Current Transformer Core

A current transformer core (*StaCtcore*) can be placed either inside a current transformer (see Section 1.3) or a combined instrument transformer (*StaCombi*). The configuration options are similar to those used for current transformers (see Section 1.3). Only the measurement location can not be changed, as it is derived from the parent transformer.

Note: Apart from sharing the same measurement location with their parent transformer, current transformer cores are completely separate models. During calculations, there is no influence of a core on its parent (or vice versa) or on any other cores in the same transformer.

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2 Calculation Models

This section describes the calculation models.

2.1 Calculation of Primary Input

Typically the input signals for primary current are automatically connected to the corresponding measurement location. The per-unit values obtained in this manner are subsequently converted into primary quantities using the nominal current of the measurement location. In contrast to most *PowerFactory* models, current transformers always provide an unbalanced output, even in balanced calculations. The following equations are used to generate the necessary primary phase currents from balanced calculations:

$$\underline{I}_{p,A} = \underline{I}_p \qquad \underline{I}_{p,B} = a \cdot \underline{I}_p \qquad \underline{I}_{p,C} = a^2 \cdot \underline{I}_p \tag{7}$$

where:

$\underline{I}_p$ is the symmetrical primary current in A

$\underline{I}_{p,X}$ is the generated primary current, phase X in A

a is the phasor rotation operator $a = e^{j120°}$

2.2 Ideal 2- or 3-Phase Current Transformer

2.2.1 Load Flow, Short-Circuit Calculation and RMS Simulation

***Wye*-connected current transformer:**

$$\underline{I}_{s,A} = \frac{1}{ratio} \cdot \underline{I}_{p,A} \tag{8}$$

$$\underline{I}_{s,B} = \frac{1}{ratio} \cdot \underline{I}_{p,B} \tag{9}$$

$$\underline{I}_{s,C} = \frac{1}{ratio} \cdot \underline{I}_{p,C} \tag{10}$$

$$3\underline{I}_{s,0} = \underline{I}_{s,A} + \underline{I}_{s,B} + \underline{I}_{s,C} \tag{11}$$

***Delta*-connected current transformer:**

$$\underline{I}_{s,A} = \frac{1}{ratio} \cdot (\underline{I}_{p,A} - \underline{I}_{p,B}) \tag{12}$$

$$\underline{I}_{s,B} = \frac{1}{ratio} \cdot (\underline{I}_{p,B} - \underline{I}_{p,C}) \tag{13}$$

$$\underline{I}_{s,C} = \frac{1}{ratio} \cdot (\underline{I}_{p,C} - \underline{I}_{p,A}) \tag{14}$$

$$3\underline{I}_{s,0} = 0 \tag{15}$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$\underline{I}_{s,X}$ is the secondary current, phase X in A (*s:I2r_X* $+ j \cdot$ *s:I2i_X*)

$3\underline{I}_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r* $+ j \cdot$ *s:I0x3i*)

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)

$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

For 2-phase, *wye*-connected current transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.1: Signals, ideal 2- or 3-phase CT, balanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ir_p	Primary Current, Real Part	p.u.	IN
Ii_p	Primary Current, Imaginary Part	p.u.	IN
I2r_A	Secondary Current, Real Part A	A	OUT
I2i_A	Secondary Current, Imaginary Part A	A	OUT
I2r_B	Secondary Current, Real Part B	A	OUT
I2i_B	Secondary Current, Imaginary Part B	A	OUT
I2r_C	Secondary Current, Real Part C	A	OUT
I2i_C	Secondary Current, Imaginary Part C	A	OUT
I0x3r	3*I0 Secondary Current, Real Part	A	OUT
I0x3i	3*I0 Secondary Current, Imaginary Part	A	OUT

Table 2.2: Signals, ideal 2- or 3-phase CT, unbalanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ir_A	Primary Current, Real Part A	p.u.	IN
Ii_A	Primary Current, Imaginary Part A	p.u.	IN
Ir_B	Primary Current, Real Part B	p.u.	IN
Ii_B	Primary Current, Imaginary Part B	p.u.	IN
Ir_C	Primary Current, Real Part C	p.u.	IN
Ii_C	Primary Current, Imaginary Part C	p.u.	IN
I2r_A	Secondary Current, Real Part A	A	OUT
I2i_A	Secondary Current, Imaginary Part A	A	OUT
I2r_B	Secondary Current, Real Part B	A	OUT
I2i_B	Secondary Current, Imaginary Part B	A	OUT
I2r_C	Secondary Current, Real Part C	A	OUT
I2i_C	Secondary Current, Imaginary Part C	A	OUT
I0x3r	3*I0 Secondary Current, Real Part	A	OUT
I0x3i	3*I0 Secondary Current, Imaginary Part	A	OUT

Table 2.3: Calculation parameters, ideal 2- or 3-phase CT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec:A	Secondary current	A
Isec:B	Secondary current	A
Isec:C	Secondary current	A
I0x3	3*I0 Secondary current	A
loading	Loading	%
Ithr	Rated Short-Time Current (1s)	A
Ithrtk	Rated Short-Time Current (Tk)	A
Ithmax	Max. Thermal equivalent Short-Circuit Current	A

The calculation parameters *c:Isec:A*, *c:Isec:B*, *c:Isec:C* and *c:I0x3* show the magnitudes of the corresponding complex secondary currents.

In load flow and RMS simulation, the loading (*c:loading*) is calculated as follows:

$$loading = max(|\underline{I}_{p,A}|, |\underline{I}_{p,B}|, |\underline{I}_{p,C}|)/I_{r,p} * 100\% \quad (16)$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A

$I_{r,p}$ is the rated primary current in A

In short-circuit calculation, the loading (*c:loading*) is calculated as follows:

$$loading = I_{th,max}/I_{thrtk} * 100\% \quad (17)$$

where:

$I_{th,max}$ is the largest thermal equivalent current from the short-circuit calculation in A

I_{thrtk} is rated thermal equivalent current (*t:Ithr*) adapted to the short-circuit duration Tk in A]

2.2.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation.

***Wye*-connected current transformer:**

$$I_{s,A} = \frac{1}{ratio} \cdot I_{p,A} \tag{18}$$

$$I_{s,B} = \frac{1}{ratio} \cdot I_{p,B} \tag{19}$$

$$I_{s,C} = \frac{1}{ratio} \cdot I_{p,C} \tag{20}$$

$$3I_{s,0} = I_{s,A} + I_{s,B} + I_{s,C} \tag{21}$$

***Delta*-connected current transformer:**

$$I_{s,A} = \frac{1}{ratio} \cdot (I_{p,A} - I_{p,B}) \tag{22}$$

$$I_{s,B} = \frac{1}{ratio} \cdot (I_{p,B} - I_{p,C}) \tag{23}$$

$$I_{s,C} = \frac{1}{ratio} \cdot (I_{p,C} - I_{p,A}) \tag{24}$$

$$3I_{s,0} = 0 \tag{25}$$

where:

$I_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$I_{s,X}$ is the secondary current, phase X in A (*s:I2r_X*)

$3I_{s,0}$ is the secondary zero sequence current in A *s:I0x3r*

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)

$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

For 2-phase, *wye*-connected current transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.4: Signals, ideal 2- or 3-phase CT, EMT simulation

Signal	Description	Unit	Type
I_A	Primary Current A	p.u.	IN
I_B	Primary Current B	p.u.	IN
I_C	Primary Current C	p.u.	IN
I2r_A	Secondary Current A	A	OUT
I2r_B	Secondary Current B	A	OUT
I2r_C	Secondary Current C	A	OUT
I0x3r	3*I0 Secondary Current	A	OUT

Table 2.5: Calculation parameters, ideal 2- or 3-phase CT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec:A	Secondary current	A
Isec:B	Secondary current	A
Isec:C	Secondary current	A
I0x3	3*I0 Secondary current	A

2.3 Detailed 2- or 3-Phase Current Transformer

2.3.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Saturation is considered with a steady state equivalent. The equivalent circuit is shown in Figure 2.1.

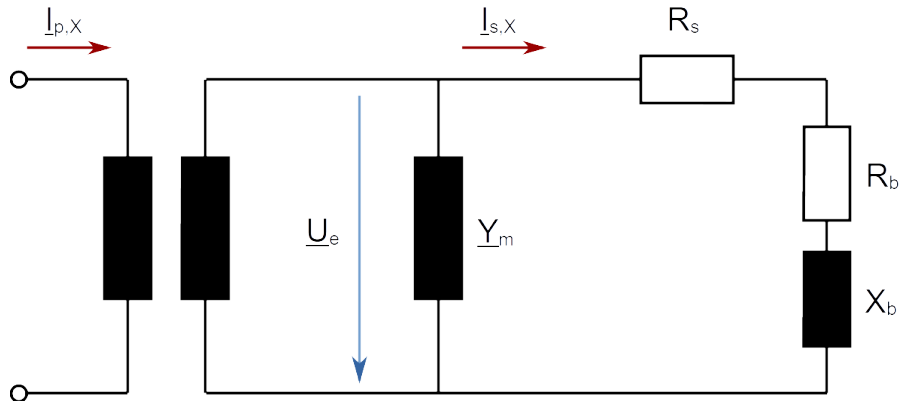


Figure 2.1: Equivalent circuit used in load flow, short-circuit calculation and RMS simulation

The complex magnetising admittance $\underline{Y}_m$ and burden $\underline{Z}_b$ are calculated as follows:

$$\underline{Y}_m = -j \cdot \frac{i_e}{Z_r} \tag{26}$$

$$\underline{Z}_b = R_b + j \cdot X_b = Z_b \cdot (\cos\varphi_b + j \cdot \sqrt{1 - (\cos\varphi_b)^2}) \tag{27}$$

where:

i_e is the excitation to rated current ratio (*e:curmg*)

Z_r is the rated burden in Ω (*t:Zb*; see also Section 1.1)

Z_b is the magnitude of the actual burden in Ω (*e:Zburd*)

$\cos\varphi_b$ is the power factor of the actual burden (*e:cosburd*)

***Wye*-connected current transformer:**

Secondary currents:

$$\underline{I}_{s,A} = \frac{1}{ratio} \cdot \underline{I}_{p,A} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{28}$$

$$\underline{I}_{s,B} = \frac{1}{ratio} \cdot \underline{I}_{p,B} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{29}$$

$$\underline{I}_{s,C} = \frac{1}{ratio} \cdot \underline{I}_{p,C} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{30}$$

$$3\underline{I}_{s,0} = \underline{I}_{s,A} + \underline{I}_{s,B} + \underline{I}_{s,C} \tag{31}$$

Excitation voltages:

$$\underline{U}_{e,A} = \underline{I}_{s,A} \cdot (R_s + \underline{Z}_b) \tag{32}$$

$$\underline{U}_{e,B} = \underline{I}_{s,B} \cdot (R_s + \underline{Z}_b) \tag{33}$$

$$\underline{U}_{e,C} = \underline{I}_{s,C} \cdot (R_s + \underline{Z}_b) \tag{34}$$

***Delta*-connected current transformer:**

Secondary currents:

$$\underline{I}_{D0} = \frac{1}{3 \cdot ratio} \cdot (\underline{I}_{p,A} + \underline{I}_{p,B} + \underline{I}_{p,C})$$

$$\underline{I}_{s,A} = \frac{1}{ratio} \cdot (\underline{I}_{p,A} - \underline{I}_{p,B}) \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + 3 \cdot \underline{Z}_b)} \quad (35)$$

$$\underline{I}_{s,B} = \frac{1}{ratio} \cdot (\underline{I}_{p,B} - \underline{I}_{p,C}) \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + 3 \cdot \underline{Z}_b)} \quad (36)$$

$$\underline{I}_{s,C} = -(\underline{I}_{s,A} + \underline{I}_{s,B}) \quad (37)$$

$$3\underline{I}_{s,0} = 0 \quad (38)$$

Excitation voltages:

$$\underline{U}_{e,A} = \underline{I}_{s,A} \cdot (R_s + 3 \cdot \underline{Z}_b) + \underline{I}_{D0} \cdot \frac{R_s}{1 + \underline{Y}_m \cdot R_s} \quad (39)$$

$$\underline{U}_{e,B} = \underline{I}_{s,B} \cdot (R_s + 3 \cdot \underline{Z}_b) + \underline{I}_{D0} \cdot \frac{R_s}{1 + \underline{Y}_m \cdot R_s} \quad (40)$$

$$\underline{U}_{e,C} = \underline{I}_{s,C} \cdot (R_s + 3 \cdot \underline{Z}_b) + \underline{I}_{D0} \cdot \frac{R_s}{1 + \underline{Y}_m \cdot R_s} \quad (41)$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$\underline{I}_{s,X}$ is the secondary current, phase X in A (*s:I2r_X*$+ j \cdot$ *s:I2i_X*)

$3\underline{I}_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r*$+ j \cdot$ *s:I0x3i*)

$\underline{U}_{e,X}$ is the excitation voltage, phase X in V

$\underline{Y}_m$ is the complex magnetising admittance in S

R_s is the secondary winding resistance in Ω (*e:Rs*)

$\underline{Z}_b$ is the complex burden in Ω

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

- $I_{tap,p}$ is the selected primary current in A (*e:ptapset*)
- $I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

For 2-phase, *wye*-connected current transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Signal connections are identical to the ideal current transformer. See Tables 2.1 and 2.2.

Table 2.6: Calculation parameters, detailed 2- or 3-phase CT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec:A	Secondary current A	A
Isec:B	Secondary current B	A
Isec:C	Secondary current C	A
I0x3	3*I0 Secondary current	A
loading	Loading	%
Zb:r	Burden, Impedance, Real Part	Ω
Zb:i	Burden, Impedance, Imaginary Part	Ω
Zb:phi	Burden, Impedance, Angle	
Zb:mag	Burden, Impedance, Magnitude	Ω
Zb:phirad	Burden, Impedance, Angle	rad
Vmax	Voltage at Accuracy Limit	V
loadsat	Saturation Loading	%
Ve:A	Excitation Voltage A	V
Ve:B	Excitation Voltage B	V
Ve:C	Excitation Voltage C	V
Ithr	Rated Short-Time Current (1s)	A
Ithrtk	Rated Short-Time Current (Tk)	A
Ithmax	Max. Thermal equivalent Short-Circuit Current	A

Only the additional calculation parameters for the detailed model are explained here. Please refer to the explanations below Table 2.3 for the calculation of the loading and the secondary currents.

The calculation parameters *c:Ve:A*, *c:Ve:B* and *c:Ve:C* show the magnitudes of the corresponding complex excitation voltages.

The saturation loading $loading_{sat}$ (*c:loadsat*) is calculated as follows:

$$loading_{sat} = max(|\underline{U}_{e,A}|, |\underline{U}_{e,B}|, |\underline{U}_{e,C}|)/V_{max} * 100\% \quad (42)$$

where:

$\underline{U}_{e,X}$ is the excitation voltage, phase X in V

V_{max} is the voltage at the accuracy limit in V (*t:Vmax*; see also Section 1.1)

2.3.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation. The saturation is considered with a non-linear magnetising admittance. The equivalent circuit is shown in Figure 2.2.

The unsaturated magnetising admittance B_m, saturated magnetising admittance $B_{m,sat}$, burden

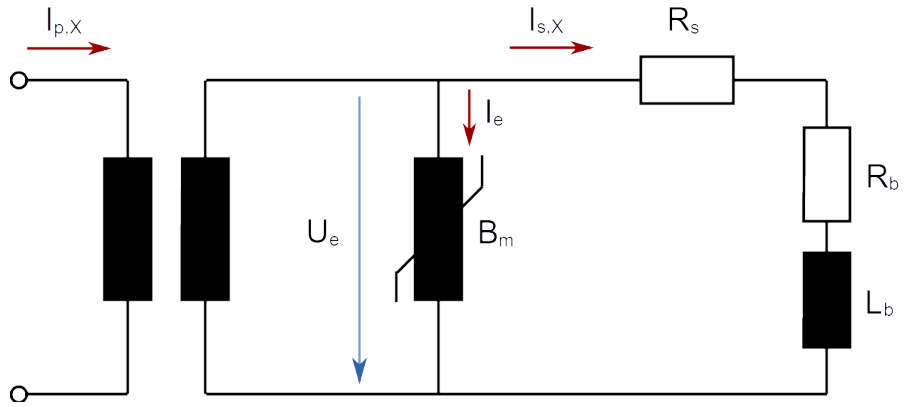


Figure 2.2: Equivalent circuit used in EMT simulation

resistance R_b and burden inductance L_b are calculated as follows:

$$B_m = \frac{i_e}{Z_r} \tag{43}$$

$$B_{m,sat} = \frac{y_{sat}}{Z_r} \tag{44}$$

$$R_b = Z_b \cdot \cos\varphi_b \tag{45}$$

$$L_b = \frac{Z_b \cdot \sqrt{1 - (\cos\varphi_b)^2)}}{2 \cdot \pi \cdot f_N} \tag{46}$$

where:

i_e is the excitation to rated current ratio in p.u. (*e:curmg*)

Z_r is the rated burden in Ω (*t:Zb*; see also Section 1.1)

y_{sat} is the saturated admittance in p.u. (*e:Bmsat*)

Z_b is the magnitude of the actual burden in Ω (*e:Zburd*)

$\cos\varphi_b$ is the power factor of the actual burden (*e:cosburd*)

f_N is the nominal frequency of the network in Hz

Wye-connected current transformer:

Secondary currents:

$$I_{s,A} = \frac{1}{ratio} \cdot I_{p,A} - I_{e,A} \tag{47}$$

$$I_{s,B} = \frac{1}{ratio} \cdot I_{p,A} - I_{e,B} \tag{48}$$

$$I_{s,C} = \frac{1}{ratio} \cdot I_{p,C} - I_{e,C} \tag{49}$$

$$3I_{s,0} = I_{s,A} + I_{s,B} + I_{s,C} \tag{50}$$

Excitation voltages:

$$U_{e,A} = I_{s,A} \cdot (R_s + R_b) + \frac{d}{dt} I_{s,A} \cdot L_b \tag{51}$$

$$U_{e,B} = I_{s,B} \cdot (R_s + R_b) + \frac{d}{dt} I_{s,B} \cdot L_b \tag{52}$$

$$U_{e,C} = I_{s,C} \cdot (R_s + R_b) + \frac{d}{dt} I_{s,C} \cdot L_b \tag{53}$$

Magnetic flux:

$$\frac{d}{dt} \Psi_A = U_{e,A} \tag{54}$$

$$\frac{d}{dt} \Psi_B = U_{e,B} \tag{55}$$

$$\frac{d}{dt} \Psi_C = U_{e,C} \tag{56}$$

Delta-connected current transformer:

Secondary currents:

$$I_{D0} = \frac{1}{3 \cdot ratio} \cdot (I_{p,A} + I_{p,B} + I_{p,C}) - (I_{e,A} + I_{e,B} + I_{e,C})$$

$$I_{s,A} = \frac{1}{ratio} \cdot I_{p,A} - I_{e,A} - I_{D0} + I_{s,B} \tag{57}$$

$$I_{s,B} = \frac{1}{ratio} \cdot I_{p,B} - I_{e,B} - I_{D0} + I_{s,C} \tag{58}$$

$$I_{s,C} = -(I_{s,A} + I_{s,B}) \tag{59}$$

$$3I_{s,0} = 0 \tag{60}$$

Excitation voltages:

$$U_{e,A} = (I_{s,A} - I_{s,B}) \cdot (R_s + 3 \cdot R_b) + \frac{d}{dt} (I_{s,A} - I_{s,B}) \cdot 3 \cdot L_b \tag{61}$$

$$U_{e,B} = (I_{s,B} - I_{s,C}) \cdot (R_s + 3 \cdot R_b) + \frac{d}{dt} (I_{s,B} - I_{s,C}) \cdot 3 \cdot L_b \tag{62}$$

$$U_{e,B} = (I_{s,B} - I_{s,C}) \cdot (R_s + 3 \cdot R_b) + \frac{d}{dt} (I_{s,B} - I_{s,C}) \cdot 3 \cdot L_b \tag{63}$$

Magnetic flux:

$$\frac{d}{dt} \Psi_A = U_{e,A} \tag{64}$$

$$\frac{d}{dt} \Psi_B = U_{e,B} \tag{65}$$

$$\frac{d}{dt} \Psi_C = U_{e,C} \tag{66}$$

where:

$I_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$I_{s,X}$ is the secondary current, phase X in A (*s:I2r_X*)

$3I_{s,0}$ is the secondary zero sequence current in A *s:I0x3r*

$U_{e,X}$ is the excitation voltage, phase X in V

$I_{e,X}$ is the excitation current, phase X in A (*c:Ie_X*; see also Section 2.6)

Ψ_X is the magnetic flux, phase X in Vs (*s:psim_X*)

R_s is the secondary winding resistance in Ω (*e:Rs*)

R_b is the burden resistance in Ω

L_b is the burden inductance in H

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)
$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

For a detailed description of the saturation characteristic, please refer to Section 2.6

For 2-phase, *wye*-connected current transformers the missing phase remains zero. In the case of a non-standard phase rotation, the primary phases are rotated accordingly.

Table 2.7: Signals and states, detailed 2- or 3-phase CT, EMT simulation

Signal	Description	Unit	Type
I_A	Primary Current A	p.u.	IN
I_B	Primary Current B	p.u.	IN
I_C	Primary Current C	p.u.	IN
I2r_A	Secondary Current A	A	OUT
I2r_B	Secondary Current B	A	OUT
I2r_C	Secondary Current C	A	OUT
I0x3r	3*I0	A	OUT
I2r_A:dt	Secondary Current, derivative A	A/s	OUT
I2r_B:dt	Secondary Current, derivative B	A/s	OUT
I2r_C:dt	Secondary Current, derivative C	A/s	OUT
I0x3r:dt	3*I0 , derivative	A/s	OUT
psim_A	Mag. Flux	Vs	STATE
psim_B	Mag. Flux	Vs	STATE
psim_C	Mag. Flux	Vs	STATE
psim_A:dt	Mag. Flux, derivative	Vs/s	d/dt
psim_B:dt	Mag. Flux, derivative	Vs/s	d/dt
psim_C:dt	Mag. Flux, derivative	Vs/s	d/dt

Table 2.8: Calculation parameters, detailed 2- or 3-phase CT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec:A	Secondary current A	A
Isec:B	Secondary current B	A
Isec:C	Secondary current C	A
I0x3	3*I0 Secondary current	A
Rs	Secondary Winding-Resistance	Ω
Lb	Burden, Inductance	H
Rb	Burden, Resistance	Ω
Bm	Mag. Inductance (1/Lm)	1/H
Bmsat	Saturated Mag. Inductance (1/Lmsat)	1/H
ksat	Saturation Exponent	
Ie_A	Excitation Current A	A
Ie_B	Excitation Current B	A
Ie_C	Excitation Current C	A

2.4 Ideal Single-Phase Current Transformer

For compatibility reasons, single-phase current transformers also provide a set of output signals with phase designation (*s:I2r_A* and *s:I2i_A*). These identical to the default signal (*s:I2r* and *s:I2i*) and are therefore not listed in the equations below.

2.4.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Connected to *A*, *B* or *C*:

$$\underline{I}_s = \frac{1}{ratio} \cdot \underline{I}_{p,X} \tag{67}$$

$$3\underline{I}_{s,0} = 0 \tag{68}$$

Connected to *I0*:

$$\underline{I}_{p,0} = \begin{cases} (\underline{I}_{p,A} + \underline{I}_{p,B} + \underline{I}_{p,C})/3 & \text{meas. location without phase 'N'} \\ \underline{I}_{p,N}/3 & \text{meas. location with phase 'N'} \\ \underline{I}_{0,Delta,X} & \text{3-W or 4-W transformer} \end{cases}$$

$$\underline{I}_s = \frac{1}{ratio} \cdot \underline{I}_{p,0} \tag{69}$$

$$3\underline{I}_{s,0} = 3 \cdot \underline{I}_s \tag{70}$$

Connected to *N*:

$$3\underline{I}_{p,0} = \begin{cases} \underline{I}_{p,A} + \underline{I}_{p,B} + \underline{I}_{p,C} & \text{meas. location without phase 'N'} \\ \underline{I}_{p,N} & \text{meas. location with phase 'N'} \end{cases}$$

$$\underline{I}_s = \frac{1}{ratio} \cdot 3\underline{I}_{p,0} \tag{71}$$

$$3\underline{I}_{s,0} = \underline{I}_s \tag{72}$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$\underline{I}_s$ is the secondary current in A (*s:I2r* $+ j \cdot$ *s:I2i*)

$3\underline{I}_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r* $+ j \cdot$ *s:I0x3i*)

$\underline{I}_{0,Delta,X}$ is the circulating current, side X in A (ElmTr3/ElmTr4; *s:I0rDelta_X* $+ j \cdot$ *s:I0iDelta_X*)

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)

$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

Table 2.9: Signals, single-phase CT, balanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ir_p	Primary Current, Real Part	p.u.	IN
Ii_p	Primary Current, Imaginary Part	p.u.	IN
I2r	Secondary Current, Real Part	A	OUT
I2i	Secondary Current, Imaginary Part	A	OUT
I2r_A	Secondary Current, Real Part	A	OUT
I2i_A	Secondary Current, Imaginary Part	A	OUT
I0x3r	3*I0, Real Part	A	OUT
I0x3i	3*I0, Imaginary Part	A	OUT

Table 2.10: Signals, ideal single-phase CT, unbalanced load flow, short-circuit calculation and RMS simulation

Signal	Description	Unit	Type
Ir_A	Primary Current, Real Part A	p.u.	IN
Ii_A	Primary Current, Imaginary Part A	p.u.	IN
Ir_B	Primary Current, Real Part B	p.u.	IN
Ii_B	Primary Current, Imaginary Part B	p.u.	IN
Ir_C	Primary Current, Real Part C	p.u.	IN
Ii_C	Primary Current, Imaginary Part C	p.u.	IN
I2r	Secondary Current, Real Part	A	OUT
I2i	Secondary Current, Imaginary Part	A	OUT
I2r_A	Secondary Current, Real Part	A	OUT
I2i_A	Secondary Current, Imaginary Part	A	OUT
I0x3r	3*I0, Real Part	A	OUT
I0x3i	3*I0, Imaginary Part	A	OUT

Table 2.11: Calculation parameters, ideal single-phase CT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec	Secondary current	A
I0x3	3*I0 Secondary current	A
loading	Loading	%
Ithr	Rated Short-Time Current (1s)	A
Ithrtk	Rated Short-Time Current (Tk)	A
Ithmax	Max. Thermal equivalent Short-Circuit Current	A

The calculation parameters *c:Isec* and *c:I0x3* show the magnitudes of the corresponding complex secondary currents.

In load flow and RMS simulation, the loading (*c:loading*) is calculated as follows (only if connected to *A*, *B* or *C*):

$$loading = |\underline{I}_{p,X}|/I_{r,p} * 100\% \tag{73}$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A

$I_{r,p}$ is the rated primary current in A

In short-circuit calculation, the loading (*c:loading*) is calculated as follows:

$$loading = I_{th,max}/I_{thrtk} * 100\% \tag{74}$$

where:

$I_{th,max}$ is the largest thermal equivalent current from the short-circuit calculation in A

I_{thrtk} is rated thermal equivalent current (*t:Ithr*) adapted to the short-circuit duration Tk in A

2.4.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation.

Connected to *A*, *B* or *C*:

$$I_s = \frac{1}{ratio} \cdot I_{p,X} \tag{75}$$

$$3I_{s,0} = 0 \tag{76}$$

Connected to *I0*:

$$I_{p,0} = \begin{cases} (I_{p,A} + I_{p,B} + I_{p,C})/3 & \text{meas. location without phase 'N'} \\ I_{p,N}/3 & \text{meas. location with phase 'N'} \\ I_{0,Delta,X} & \text{3-W or 4-W transformer} \end{cases}$$

$$I_s = \frac{1}{ratio} \cdot I_{p,0} \tag{77}$$

$$3I_{s,0} = 3 \cdot I_s \tag{78}$$

Connected to *N*:

$$3I_{p,0} = \begin{cases} I_{p,A} + I_{p,B} + I_{p,C} & \text{meas. location without phase 'N'} \\ I_{p,N} & \text{meas. location with phase 'N'} \end{cases}$$

$$I_s = \frac{1}{ratio} \cdot 3I_{p,0} \tag{79}$$

$$3I_{s,0} = I_s \tag{80}$$

where:

$I_{p,X}$ is the primary current, phase X in A (see Section 2.1)

I_s is the secondary current in A (*s:I2r*)

$3I_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r*)

$I_{0,Delta,X}$ is the circulating current, side X in A (ElmTr3/ElmTr4; *s:I0Delta_X*)

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)

$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

Table 2.12: Signals, ideal single-phase CT, EMT simulation

Signal	Description	Unit	Type
I_A	Primary Current, A	p.u.	IN
I_B	Primary Current, B	p.u.	IN
I_C	Primary Current, C	p.u.	IN
I2r	Secondary Current	A	OUT
I2r_A	Secondary Current	A	OUT
I0x3r	3*I0	A	OUT

Table 2.13: Calculation parameters, ideal single-phase CT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec	Secondary current	A
I0x3	3*I0 Secondary current	A

2.5 Detailed Single-Phase Current Transformer

2.5.1 Load Flow, Short-Circuit Calculation and RMS Simulation

Saturation is considered with a steady state equivalent. The equivalent circuit is identical to the 2- or 3-phase current transformer. Please refer to Figure 2.1.

Connected to *A*, *B* or *C*:

Secondary currents:

$$\underline{I}_s = \frac{1}{ratio} \cdot \underline{I}_{p,X} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{81}$$

$$3\underline{I}_{s,0} = 0 \tag{82}$$

Excitation voltage:

$$\underline{U}_e = \underline{I}_s \cdot (R_s + \underline{Z}_b) \tag{83}$$

Connected to *I0*:

Secondary currents:

$$\underline{I}_{p,0} = \begin{cases} (\underline{I}_{p,A} + \underline{I}_{p,B} + \underline{I}_{p,C})/3 & \text{meas. location without phase 'N'} \\ \underline{I}_{p,N}/3 & \text{meas. location with phase 'N'} \\ \underline{I}_{0,Delta,X} & \text{3-W or 4-W transformer} \end{cases}$$

$$\underline{I}_s = \frac{1}{ratio} \cdot \underline{I}_{p,0} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{84}$$

$$3\underline{I}_{s,0} = 3 \cdot \underline{I}_s \tag{85}$$

Excitation voltage:

$$\underline{U}_e = 3 \cdot \underline{I}_s \cdot (R_s + \underline{Z}_b) \tag{86}$$

Connected to *N*:

Secondary currents:

$$3\underline{I}_{p,0} = \begin{cases} \underline{I}_{p,A} + \underline{I}_{p,B} + \underline{I}_{p,C} & \text{meas. location without phase 'N'} \\ \underline{I}_{p,N} & \text{meas. location with phase 'N'} \end{cases}$$

$$\underline{I}_s = \frac{1}{ratio} \cdot 3\underline{I}_{p,0} \cdot \frac{1}{1 + \underline{Y}_m \cdot (R_s + \underline{Z}_b)} \tag{87}$$

$$3\underline{I}_{s,0} = \underline{I}_s \tag{88}$$

Excitation voltage:

$$\underline{U}_e = \underline{I}_s \cdot (R_s + \underline{Z}_b) \tag{89}$$

where:

$\underline{I}_{p,X}$ is the primary current, phase X in A (see Section 2.1)

$\underline{I}_s$ is the secondary current in A (*s:I2r* $+ j \cdot$ *s:I2i*)

$3\underline{I}_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r* $+ j \cdot$ *s:I0x3i*)

$\underline{I}_{0,Delta,X}$ is the circulating current, side X in A (ElmTr3/ElmTr4; *s:I0rDelta_X* $+ j \cdot$ *s:I0iDelta_X*)

$\underline{U}_e$ is the excitation voltage in V

$\underline{Y}_m$ is the complex magnetising admittance in [S]

R_s is the secondary winding resistance in Ω (*e:Rs*)

$\underline{Z}_b$ is the complex burden in Ω

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

$I_{tap,p}$ is the selected primary current in A (*e:ptapset*)
$I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

Signal connections are identical to the ideal current transformer. See Tables 2.9 and 2.10.

Table 2.14: Calculation parameters, detailed single-phase CT, load flow, short-circuit calculation and RMS simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec	Secondary current	A
I0x3	3*I0 Secondary current	A
loading	Loading	%
Zb:r	Burden, Impedance, Real Part	Ω
Zb:i	Burden, Impedance, Imaginary Part	Ω
Zb:phi	Burden, Impedance, Angle	
Zb:mag	Burden, Impedance, Magnitude	Ω
Zb:phirad	Burden, Impedance, Angle	rad
Vmax	Voltage at Accuracy Limit	V
loadsat	Saturation Loading	%
Ve	Excitation Voltage	V
Ithr	Rated Short-Time Current (1s)	A
Ithrtk	Rated Short-Time Current (Tk)	A
Ithmax	Max. Thermal equivalent Short-Circuit Current	A

Only the additional calculation parameters for the detailed model are explained here. Please refer to the explanations below Table 2.11 for the calculation of the loading and the secondary currents.

The calculation parameter *c:Ve* shows the magnitude of the complex excitation voltage.

The saturation loading $loading_{sat}$ (*c:loadsat*) is calculated as follows:

$$loading_{sat} = |\underline{U}_e|/V_{max} * 100\% \tag{90}$$

where:

$\underline{U}_e$ is the excitation voltage in V

V_{max} is the voltage at the accuracy limit in V (*t:Vmax*; see also Section 1.1)

2.5.2 EMT Simulation

The model for EMT simulation uses instantaneous values instead of the complex phasors used in load flow, short-circuit calculation and RMS simulation. The saturation is considered with a non-linear magnetising admittance. The equivalent circuit is identical to the 2- or 3-phase current transformer. Please refer to Figure 2.2.

Connected to *A*, *B* or *C*:

Secondary currents:

$$I_s = \frac{1}{ratio} \cdot I_{p,X} - I_e \tag{91}$$

$$3I_{s,0} = 0 \tag{92}$$

Excitation voltage:

$$U_e = I_s \cdot (R_s + R_b) + \frac{d}{dt} I_s \cdot L_b \tag{93}$$

Magnetic flux:

$$\frac{d}{dt}\Psi = U_e \tag{94}$$

Connected to *I0*:

Secondary currents:

$$I_{p,0} = \begin{cases} (I_{p,A} + I_{p,B} + I_{p,C})/3 & \text{meas. location without phase 'N'} \\ I_{p,N}/3 & \text{meas. location with phase 'N'} \\ I_{0,Delta,X} & \text{3-W or 4-W transformer} \end{cases}$$

$$I_s = \frac{1}{ratio} \cdot I_{p,0} - \frac{I_e}{3} \tag{95}$$

$$3I_{s,0} = 3 \cdot I_s \tag{96}$$

Excitation voltage:

$$U_e = 3I_{s,0} \cdot (R_s + R_b) + \frac{d}{dt} 3I_{s,0} \cdot L_b \tag{97}$$

Magnetic flux:

$$\frac{d}{dt}\Psi = U_e \tag{98}$$

Connected to *N*:

$$3I_{p,0} = \begin{cases} I_{p,A} + I_{p,B} + I_{p,C} & \text{meas. location without phase 'N'} \\ I_{p,N} & \text{meas. location with phase 'N'} \end{cases}$$

$$I_s = \frac{1}{ratio} \cdot 3I_{p,0} \tag{99}$$

$$3I_{s,0} = I_s \tag{100}$$

Excitation voltage:

$$U_e = I_s \cdot (R_s + R_b) + \frac{d}{dt} I_s \cdot L_b \tag{101}$$

Magnetic flux:

$$\frac{d}{dt}\Psi = U_e \tag{102}$$

where:

$I_{p,X}$ is the primary current, phase X in A (see Section 2.1)

I_s is the secondary current in A (*s:I2r*)

$3I_{s,0}$ is the secondary zero sequence current in A (*s:I0x3r*)

$I_{0,Delta,X}$ is the circulating current, side X in A (ElmTr3/ElmTr4; *s:I0Delta_X*)

U_e is the excitation voltage in V

I_e is the excitation current in A (*c:Ie*; see also Section 2.6)

Ψ is the magnetic flux, phase X in Vs (*s:psim*)

R_s is the secondary winding resistance in Ω (*e:Rs*)

R_b is the burden resistance in Ω

L_b is the burden inductance in H

$ratio$ is the transmission ratio $ratio = I_{tap,p}/I_{tap,s}$ (*c:ratio*), where:

- $I_{tap,p}$ is the selected primary current in A (*e:ptapset*)
- $I_{tap,s}$ is the selected secondary current in A (*e:stapset*)

Table 2.15: Signals, detailed single-phase CT, EMT simulation

Signal	Description	Unit	Type
I_A	Primary Current, A	p.u.	IN
I_B	Primary Current, B	p.u.	IN
I_C	Primary Current, C	p.u.	IN
I2r	Secondary Current	A	OUT
I2r_A	Secondary Current	A	OUT
I0x3r	3*I0	A	OUT
I2r:dt	Secondary Current, derivative	A/s	OUT
I2r_A:dt	Secondary Current, derivative	A/s	OUT
I0x3r:dt	3*I0, derivative	A/s	OUT
psim	Mag. Flux	Vs	STATE
psim:dt	Mag. Flux, derivative	Vs/s	d/dt

Table 2.16: Calculation parameters, detailed single-phase CT, EMT simulation

Parameter	Description	Unit
ratio	Ratio	
seccon	Connection, secondary	
Isec	Secondary current	A
I0x3	3*I0 Secondary current	A
Rs	Secondary Winding-Resistance	Ω
Lb	Burden, Inductance	H
Rb	Burden, Resistance	Ω
Bm	Mag. Inductance (1/Lm)	1/H
Bmsat	Saturated Mag. Inductance (1/Lmsat)	1/H
ksat	Saturation Exponent	
Ie	Excitation Current	A

2.6 Saturation Characteristic

This section describes the saturation characteristics available for EMT simulation. Please refer to Figure 2.2 and corresponding the equations for the calculation of B_m and $B_{m,sat}$.

2.6.1 Piecewise linear saturation

$$I_{e,X} = \begin{cases} I_{knee} + B_{m,sat} \cdot (\Psi_X - \Psi_{knee}) & \text{if } \Psi_X > \Psi_{knee} \\ B_m \cdot \Psi_X & \text{if } -\Psi_{knee} \leq \Psi_X \leq \Psi_{knee} \\ -I_{knee} + B_{m,sat} \cdot (\Psi_X + \Psi_{knee}) & \text{if } \Psi_X < -\Psi_{knee} \end{cases} \quad (103)$$

where:

$I_{e,X}$ is the excitation current, phase X in A

Ψ_X is the magnetic flux, phase X in Vs

Ψ_{knee} is the knee-point flux in Vs

I_{knee} is the knee-point current in A

B_m is the unsaturated magnetising admittance in 1/H

$B_{m,sat}$ is the saturated magnetising admittance in 1/H

The relevant parameters at knee-point are calculated as follows:

$$\Psi_{knee} = \frac{\sqrt{2} \cdot V_s}{2 \cdot \pi \cdot f_N} \quad (104)$$

$$I_{knee} = B_m \cdot \Psi_{knee} \quad (105)$$

where:

V_s is the saturation voltage in V (*e:Vs*)

f_N is the nominal frequency of the network in Hz

B_m is the unsaturated magnetising admittance in 1/H

2.6.2 Polynomial saturation

$$I_{e,X} = \begin{cases} I_{knee} + B_{m,sat} \cdot (\Psi_X - \Psi_{knee}) & \text{if } \Psi_X > \Psi_{knee} \\ B_m \cdot \Psi_X \cdot \left(1 + \left|\frac{\Psi_X}{\Psi_0}\right|^{k_{sat}}\right) & \text{if } -\Psi_{knee} \leq \Psi_X \leq \Psi_{knee} \\ -I_{knee} + B_{m,sat} \cdot (\Psi_X + \Psi_{knee}) & \text{if } \Psi_X < -\Psi_{knee} \end{cases} \quad (106)$$

where:

$I_{e,X}$ is the excitation current, phase X in A

Ψ_X is the magnetic flux, phase X in Vs

Ψ_{knee} is the knee-point flux in Vs

Ψ_0 is the nominal flux in Vs

I_{knee} is the knee-point current in A

k_{sat} is the saturation exponent (*e:ksat*)

B_m is the unsaturated magnetising admittance in 1/H

$B_{m,sat}$ is the saturated magnetising admittance in 1/H

The relevant parameters at knee-point and the nominal flux are calculated as follows:

$$\Psi_{knee} = \frac{\sqrt{2} \cdot V_s}{2 \cdot \pi \cdot f_N} \cdot \frac{k_{sat}+1}{k_{sat}} \tag{107}$$

$$k_0 = \frac{-\ln\left(\frac{\left(\frac{B_{m,sat}}{B_m}\right)-1}{k_{sat}+1}\right)}{k_{sat}}$$

$$\Psi_0 = \Psi_{knee} \cdot e^{k_0} \tag{108}$$

$$I_{knee} = B_m \cdot \Psi_{knee} \cdot \left(1 + \left|\frac{\Psi_{knee}}{\Psi_0}\right|^{k_{sat}}\right) \tag{109}$$

where:

V_s is the saturation voltage in V (*e:Vs*)

f_N is the nominal frequency of the network in Hz

B_m is the unsaturated magnetising admittance in 1/H

$B_{m,sat}$ is the saturated magnetising admittance in 1/H

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